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Full Length Article

# Novel method to study saturation of silicon photomultiplier in scintillation detector

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ABSTRACT

Signal saturation in the Silicon PhotoMultiplier (SiPM) often causes a serious problem in using a detector equipped with SiPMs for scintillator readout. Conventionally, SiPM signal saturation is measured through direct injection of visible light pulses to the SiPM. However, the method works poorly with the scintillation detector because the effects of the time constant of the emission of scintillation light are not included. Here, we present a novel method to measure the precise degree of SiPM signal saturation in scintillation detectors. The key difference from the conventional method is to use UV light pulses, as opposed to visible light ones, to inject and excite scintillation light in the detector. It allows us to directly measure the degree of saturation in situ taking into account all the major effects in the detector, such as the time constant of the scintillation emission. We also develop a new model for SiPM signal saturation and validate it with the measured degree of saturation with the above-mentioned new method.

## 1. Introduction

Silicon PhotoMultipliers (SiPMs) are widely used in experiments in various fields. Indeed, the SiPM has many advantages, most notably a high gain with a bias voltage under 100 V, compact size, and magnetic-field hardness, to name a few. It has, though, one significant disadvantage of a non-linear response for high-intensity light input. When a large number of photons exceeding a certain threshold are injected into a SiPM, the output signal of the SiPM will be saturated due to a limited number of pixels.

The response of the SiPM is non-linear, thus called the saturation curve, and is usually measured with direct injection of fast visible-light pulses (∼400 nm) to a SiPM [1]. However, this conventional method does not measure the non-linear response with the SiPM in the scintillation detector because the method does not take into account the effect of the time constant on the order of a few nanoseconds of the scintillation light emission, which is not negligible compared with the recovery time, the order of a few to tens of nanoseconds, of the SiPM pixel. In the light of this situation, we have proposed a novel method to measure the SiPM saturation in the scintillation detector with scintillation light [2]. This method excites scintillation light by injecting fast UV pulses, as opposed to conventional visible-light ones, to the scintillator. The saturation curve measured with this method takes into account the effect of the pixel recovery during the scintillation emission and thus can directly be used for correcting saturation effects. Our previous and preliminary study [2] lacked precise measurements with experiments and detailed model validation. Here, we evaluate the new method in detail, using a new, improved measurement setup and further tuned model.

## 2. Saturation measurement

This section describes the experimental setup, the analysis procedure, and the results of our calibration and saturation measurements.

### 2.1. Setup

Fig. 1 shows the experimental setup for our saturation measurement. The injected UV light is focused by a convex lens, passes through a variable ND filter and a band-pass filter, and then enters a scintillator. The variable ND filter controls the light intensity, and the band-pass filter allows only the light from 250 nm to 270 nm to pass through. The scintillation light is excited by the injected UV light, and the SiPM detects the scintillation light. The photomultiplier tube (PMT), which has a much wider dynamic range than a SiPM, is used to measure the relationship between the excited scintillation light and the injecting

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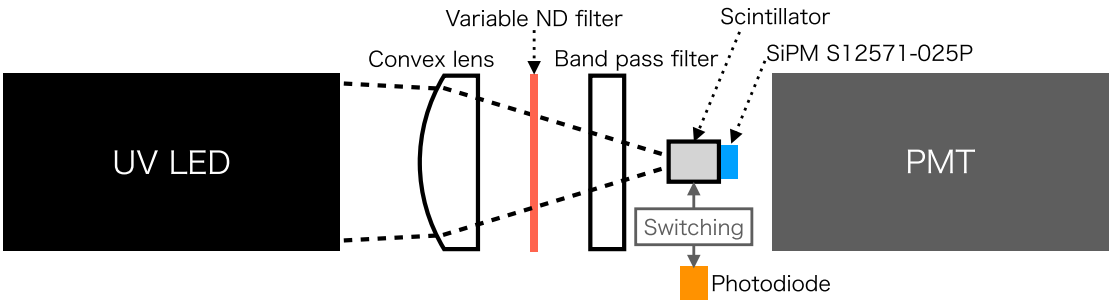


**Fig. 1.** Setup of the experiments for SiPM signal saturation measurements using UV light.

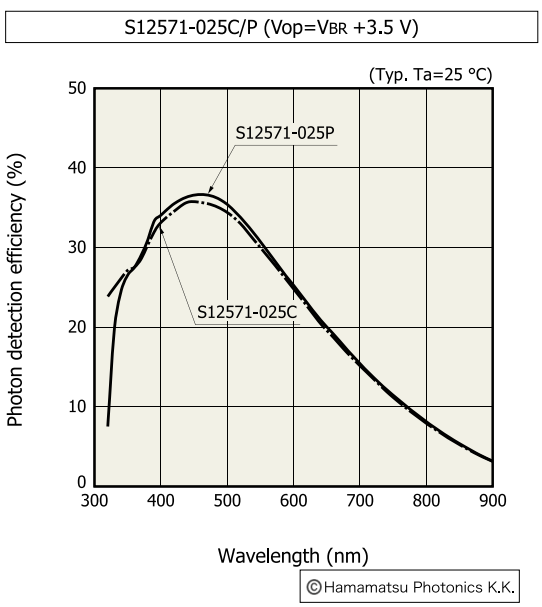


**Fig. 2.** Correlation between photon detection efficiency and wavelength for MPPC S12571-025P/C.
*Source:* This figure provided by Hamamatsu Photonics K.K. [3].

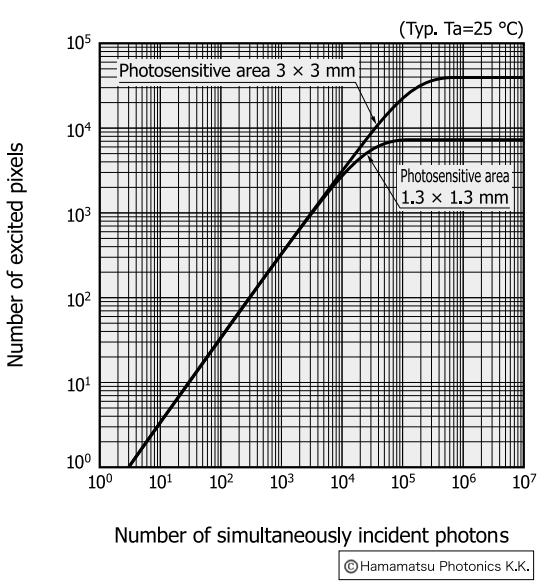


**Fig. 3.** SiPM saturation curve measured with visible-light pulses.
*Source:* The figure provided by Hamamatsu Photonics K.K. [4].

**Table 1**
Properties of EJ-200 [7].

| | |
|---|---|
| Light output | 64% |
| Rise time | 0.9 ns |
| Decay time | 2.1 ns |
| Wavelength of maximum emission | 425 nm |
| Light Attenuation length | 380 cm |

light intensity. The temperature of the SiPM is monitored by a Pt-100 temperature sensor placed behind the SiPM.

A sub-nanosecond UV pulse LED with a wavelength of 255 nm (PicoQuant PLS-255 [5]) is used in this study. For comparison with the conventional method with a visible-light pulse, a fast-pulse laser with a pulse width of 35 ps and a wavelength of 375 nm (ALS PiL037X [6]) is also used. A scintillator (Eljen, EJ-200 [7]) of 2 mm × 2 mm × 2 mm is paired with a SiPM with an active area of 1 mm × 1 mm and a pixel pitch of 25 μm (Hamamatsu MPPC S12571-025P [8]). The properties of EJ-200 are summarized in Table 1. The SiPM is operated with a voltage of 68.5 V as recommended by the manufacturer.

The sensitivity of the SiPM we used (S12571-025P) is rapidly decreasing on the wavelength shorter than 330 nm as shown in Fig. 2, from which we expect no sensitivity on UV light with a wavelength shorter than 300 nm. Therefore, if we inject such short-wavelength UV light to a scintillator, including a standard plastic one, we should be able to measure the degree of saturation of scintillation in the scintillator without direct photon injection to the SiPM.

### 2.2. Measurement

For example, Fig. 3, supplied by Hamamatsu Photonics K.K., shows the typical saturation curve measured with the conventional method with visible-light pulses [4]. We want to make a similar figure and saturation curve using UV light.

In this study, the number of simultaneously incident photons is defined as $N_{seed}$, which is equal to the expected number of the photoelectrons under an assumption of no saturation, where the correlated noise of the SiPM, such as the cross-talk and after-pulse, is taken into account. The number of the excited pixels is defined as $N_{det}$, which is equal to the number of the photoelectrons detected by the SiPM. Scanning the light intensities yields the saturation curve.

In this measurement, we estimate $N_{seed}$ from the amount of the input light measured with a photodiode and measure $N_{det}$ with the SiPM. In practice, both the parameters are first measured as current because it is impossible to analyze the whole range in units of the number of photoelectrons, and the current is then converted to the number of photoelectrons, where the calibration factor measured at low light intensities is taken into account.

The new method switches the following three types of measurements as illustrated in Fig. 4.

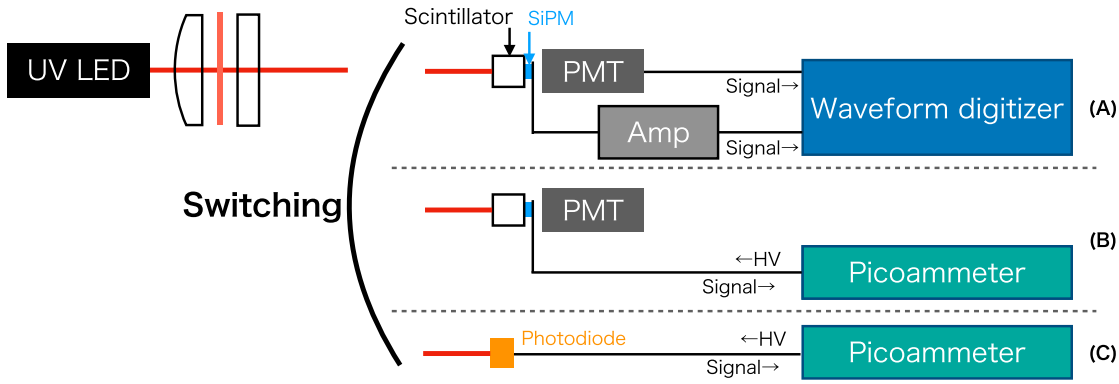


**Fig. 4.** Readout methods for the saturation measurement.

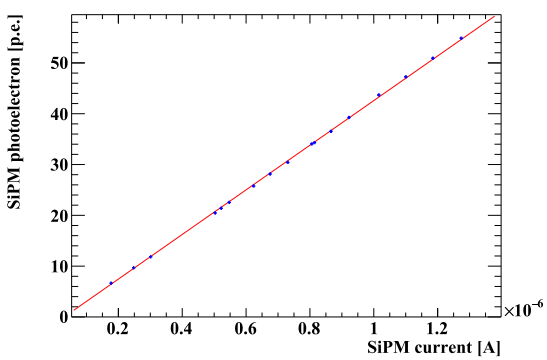


**Fig. 5.** Correlation between the SiPM current (B) and the number of photoelectrons (A). Errors are standard errors ($\frac{\sigma}{\sqrt{N}}$) of the measured current data. Measured values include pedestal offsets, then the fitting function does not pass the origin. These offsets are corrected during calibration.

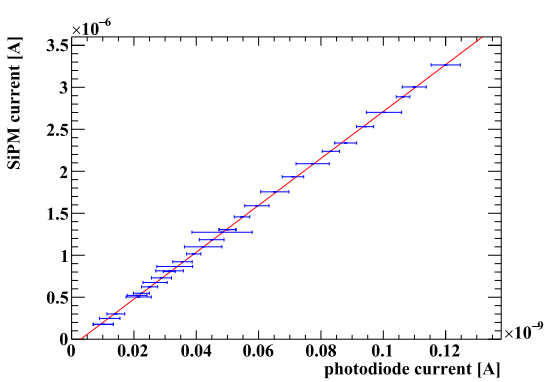


**Fig. 6.** Correlation between the photodiode current (C) and SiPM current (B). Handling of errors and offsets are the same as in Fig. 5.

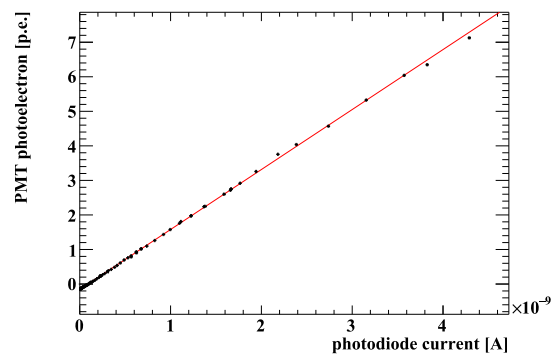


**Fig. 7.** Linearity of the scintillation light emission measured by the PMT (A) as a function of the photodiode current (C). Handling of errors and offsets are the same as in Fig. 5.

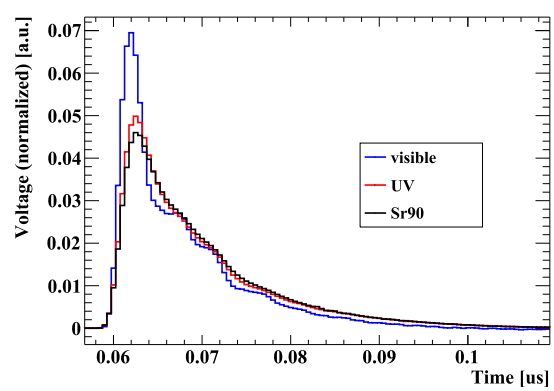


**Fig. 8.** Comparison of the signal waveform with the UV light (red), visible light (blue), and $\beta$-ray from $^{90}$Sr (black).

(A) The signal from the SiPM is measured by a waveform digitizer, of which the dynamic range for the signal input is limited to a low light-intensity range. The number of the photoelectrons is then estimated. At the same time, the signal from the PMT is also monitored by the waveform digitizer over the whole light-intensity range.

(B) The signal from the SiPM is read by a picoammeter over the whole light-intensity range

(C) The light intensity is monitored by a photodiode over the whole light-intensity range

In measurement (A), the signal from the SiPM is counted as the number of photoelectrons. For the measurement, the waveform digitizer DRS4 manufactured by the Paul Scherrer Institute [9] is used, which integrates signal waveforms to obtain the amount of charge and then convert it to the number of photoelectrons, using the single photoelectron charge determined with the dark noise spectrum. Since the dynamic range of the DRS4 is limited, this measurement is performed only when the intensity of incident light is low. The DRS4 also measures the signal from the PMT to measure the linearity of the scintillation emission. The amount of the scintillation light injected into the PMT is very small due to limited geometrical acceptance, and so is the signal from the PMT, which always falls within the acceptable range of the DRS4 regardless of the incident light intensity.

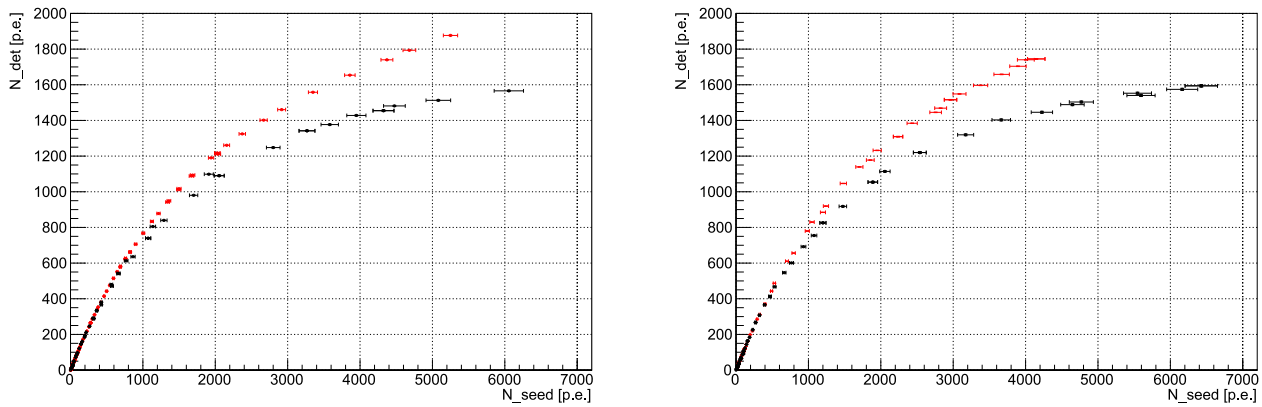


**Fig. 9.** Saturation curves measured with (red) UV light and (black) visible light. $N_{seed}$ is the expected number of the photoelectrons under an assumption of no saturation, and $N_{det}$, is the number of the photoelectrons detected by the SiPM (see Section 2.2 for detail). The number of pixels of SiPM S12571-025P is 1600. The same measurement is made with two SiPM samples. Error calculations are based on the numbers of the photo-electrons, and the standard deviations of the measured current data are displayed.

In measurement (B), the signal from the SiPM is measured as current by a picoammeter (Tektronix model 6487 [10]). Unlike the DRS4, the picoammeter has a sufficiently wide dynamic range to cover the whole expected intensity range of incident light.

In measurement (C), the UV light intensity is monitored by a UV-sensitive photodiode (Hamamatsu S12689-02), which has a large dynamic range enough to monitor the whole expected intensity range of incident light. The photodiode outputs the signal to the picoammeter, which measures it as current.

The number of events per measurement using the DRS4 is 20,000, while that using the picoammeter is 300. Because the data acquisition rate of the picoammeter is much lower than that of the DRS4, we choose a small number of events to avoid increasing measurement time.

The temperature of the system was stable within 24.5–25.0 degrees during the measurements.

### 2.3. Calibration

The currents of the SiPM and photodiode are converted to the number of photoelectrons using the linear response of the SiPM measured at low light intensity. Fig. 5 shows the correlation between the current and the number of photoelectrons. The current of the SiPM signal is converted to $N_{det}$ by applying the slope and subtracting the offset using the linear fitting function. The current of the photodiode is also converted to $N_{seed}$, and the result is shown in Fig. 6.

We find a clear linear relationship between the photodiode current and the number of photoelectrons measured by the PMT (Fig. 7), which we hoped to be the case but needed to be experimentally confirmed. This result demonstrates that the scintillation light emission linearly increases with the injected UV light input.

Fig. 8 shows a comparison between the waveforms for direct injection of visible-light pulses to the SiPM and the scintillation induced by $\beta$-rays from $^{90}$Sr and UV-light pulses. The waveforms with the UV-light pulses and $\beta$-rays are nearly identical,[1] while the waveform with the visible light is sharper. We consider this result to be supporting evidence for scintillation emission excited by UV-light pulses.

[1] There is a slight difference in the excited scintillation between the UV-light origin and $\beta$-ray origin. The UV-light pulses interact at the surface of the scintillator, whereas the $\beta$-rays interact inside the scintillator. As a result, the time constants of the resultant scintillation emissions are slightly different. Both cases are possible in experiments, and we expect that the difference causes no significant effect on this experiment.

### 2.4. Results

Fig. 9 shows the measured saturation curves (in red) for two SiPM samples. For comparison, we overlay in black the measurements in the same setup but conventionally using pulses of visible light as opposed to UV light as in our method. The figure shows a marked difference between the saturation curves with UV light and visible light. In the former, i.e., our new method, clearly visible is a saturation recovery, which is the process where some of the fired pixels recover during the period of emitting scintillation light and become ready to receive the next scintillation photons. This result demonstrates that the effect of the time constant of scintillation-light emission has a significant effect on the saturation curve.

## 3. Model

Models of SiPM signal saturation have been proposed in relation to cross-talk between pixels, an extra electron avalanche in a pixel that occurs after a photon injection (after-pulse), and the response of a recovering SiPM [1,11]. However, no previously-published models consider all the effects in the saturation process, such as the time constant of the scintillation light emission, recovery time of the SiPM, cross-talk, and after-pulse.

Here, we develop a new model of saturation, which takes account of all the four effects listed above. The model calculates the number of detected photons relative to the number of incident photons for each event and derives the saturation curve. The calculation procedure in the new model is summarized as follows:

1. Calculating the number of the incident photons $\Delta I$ at each time step within one event.
2. Calculating the number of the detected photons $N_{det}$ at each time step.
3. Summing $N_{det}$ over all the time steps.

### 3.1. Scintillation light emission

The scintillation light emission has a time constant. In the case of EJ-200, the rise time is $\tau_{rise}$ = 0.9 ns and the decay time is $\tau_{decay}$ = 2.1 ns. The number of incident photons as a function of time $I(t)$ for scintillation light emission is conventionally described as:

$$I(t) = I_0 \times (e^{-t/\tau_{decay}} - e^{-t/\tau_{rise}}), \quad (1)$$

where $I_0$ is the factor value depending on the input light intensity. In this modeling, $I_0$ depends on time due to the time dependence of UV-light pulses of the UV-LED PLS-255 used in Section 2, $I_0(t)$. The full width at half maximum (FWHM) of a light pulse of the UV-LED is about

0.5 ns. The time profile of the light pulse is approximated by a Gaussian function with $\sigma_0 = \frac{0.5}{2.35}$ ns and $\mu_0 = 1.5$ ns:

$$I_0(t) = I_0 e^{-\frac{(t-\mu_0)^2}{2\times\sigma_0^2}}, \tag{2}$$

where the Gaussian function is shifted by 1.5 ns in order that analysis can be started from the beginning of the pulse.

The time range for the calculation of the incident photons and detected photons is set from 0 to 100 ns. The time step is divided into 0.1-ns intervals. The number of the incident photons $\Delta I(t)$ is calculated at this time step according to:

$$\Delta I(t) = \sum_{t_i=0}^{t_i=t} \int_t^{t+\Delta t} I_0 e^{-\frac{(t_i-\mu_0)^2}{2\times\sigma_0^2}} \Delta t \times (e^{-(u-t_i)/\tau_{decay}} - e^{-(u-t_i)/\tau_{rise}}) du, \tag{3}$$

where $\Delta t$ is the interval of the time step (0.1 ns).[2]

### 3.2. Recovery time for a SiPM pixel

When incident light has a time distribution, there is a possibility that a photon is injected again during or after the recovery of the fired pixel, and this effect should be considered. For an injection during the recovery, an electron avalanche occurs at a reduced gain. The over-voltage $V_{over}$ of the fired pixel drops to zero and recovers exponentially with a time constant, as described with

$$V_{over}(t) = V_0(1 - e^{-t/\tau_{rec}}), \tag{4}$$

where $t$ is the time from the initial pixel trigger, $\tau_{rec}$ is the recovery time of the pixel, and $V_0$ is the initial over-voltage with no photon injection. While the pixel is recovering, we assume that the PDE $\varepsilon_{rec}(t)$ and gain $q_{rec}(t)$ depend on $V_{over}(t)$ almost linearly:

$$\varepsilon_{rec}(t) = \varepsilon_0 \frac{V_{over}(t)}{V_0} \tag{5}$$

$$q_{rec}(t) = q_0 \frac{V_{over}(t)}{V_0}, \tag{6}$$

where $\varepsilon_0$ and $q_0$ are the photon detection efficiency (PDE) and gain, respectively, at the over-voltage of $V_0$. In this model, $\varepsilon_0 = 1$ and $q_0 = 1$ for simplicity (without sacrificing generality).

### 3.3. Calculation of the number of the detected photons

Calculations of $N_{seed}(t)$ and $N_{det}(t)$ are performed for each time step. The number of incident photons, $\Delta I(t)$, is calculated at each step, and the number of the detected photons is calculated using $\Delta I(t)$. The number of the fired pixels at $t$, $N_t$, and that of the pixels with no photon injection until $t$, $N_{empty}$, are stored at each step. For example, at the first time step of $t = t_0 = 0$ ns,

$$N_{seed}(t_0) = \Delta I(t_0) \tag{7}$$

$$N_{det}(t_0) = \Delta I(t_0) \tag{8}$$

$$N_{t_0}(t_0) = \Delta I(t_0) \tag{9}$$

$$N_{empty}(t_0) = N_{pixel} - \Delta I(t_0), \tag{10}$$

where $N_{pixel}$ is the number of pixels of the SiPM.

At the second time step of $t = t_1 = 0.1$ ns, $\Delta I(t_1)$ is injected to the SiPM. The empty pixels ($N_{empty}$) can detect then injected photons. For the fired pixels ($N_{t_0}$), the effect of pixel recovery must be taken into account. The number of the pixels that detect photons at $t_1$, $N_{t_1}$, is also stored in addition to $N_{t_0}$. We update $N_{empty}$ and $N_{t_0}$ by subtracting the number of photons injected at $t_1$. Then, at $t = t_1 = 0.1$ ns,

$$N_{seed}(t_1) = \Delta I(t_1) \times \frac{N_{empty}(t_0)}{N_{pixel}} \tag{11}$$

$$+ \Delta I(t_1) \times \frac{N_{t_0}(t_0)}{N_{pixel}} \times (1 - e^{-(t_1-t_0)/\tau_{rec}}) \tag{12}$$

$$N_{det}(t_1) = \Delta I(t_1) \times \frac{N_{empty}(t_0)}{N_{pixel}} \tag{13}$$

$$+ \Delta I(t_1) \times \frac{N_{t_0}(t_0)}{N_{pixel}} \times (1 - e^{-(t_1-t_0)/\tau_{rec}})^2 \tag{14}$$

$$N_{t_0}(t_1) = N_{t_0}(t_0) - \Delta I(t_1) \times \frac{N_{t_0}(t_0)}{N_{pixel}} \times (1 - e^{-(t_1-t_0)/\tau_{rec}}) \tag{15}$$

$$N_{t_1}(t_1) = N_{seed}(t_1) \tag{16}$$

$$N_{empty}(t_1) = N_{empty}(t_0) - \Delta I(t_1) \times \frac{N_{empty}(t_0)}{N_{pixel}}, \tag{17}$$

where $\frac{N_{empty}(t_0)}{N_{pixel}}$ is the probability that $\Delta I(t_1)$ is injected to empty pixels, and $\frac{N_{t_0}}{N_{pixel}}$ is the probability that $\Delta I(t_1)$ is injected to the pixels fired at $t_0$.

The recursive relation with $t$ and $t' = t - 0.1$ is given by:

$$\Delta I(t) = \sum_{t_i=0}^{t_i=t} I_0 e^{-\frac{(t_i-\mu_0)^2}{2\times\sigma_0^2}} \Delta t \tag{18}$$

$$\times \int_t^{t+\Delta t} (e^{-(u-t_i)/\tau_{decay}} - e^{-(u-t_i)/\tau_{rise}}) du \tag{19}$$

$$N_{seed}(t) = \Delta I(t) \times \frac{N_{empty}(t')}{N_{pixel}} \tag{20}$$

$$+ \sum_{t_i=0}^{t_i=t'} \Delta I(t) \times \frac{N_{t_i}(t')}{N_{pixel}} \times (1 - e^{-(t-t_i)/\tau_{rec}}) \tag{21}$$

$$N_{det}(t) = \Delta I(t) \times \frac{N_{empty}(t')}{N_{pixel}} \tag{22}$$

$$+ \sum_{t_i=0}^{t_i=t'} \Delta I(t) \times \frac{N_{t_i}(t')}{N_{pixel}} \times (1 - e^{-(t-t_i)/\tau_{rec}})^2 \tag{23}$$

$$N_{t_i}(t) = N_{t_i}(t') - \Delta I(t) \times \frac{N_{t_i}(t')}{N_{pixel}} \times (1 - e^{-(t-t_i)/\tau_{rec}}) \tag{24}$$

$$N_t(t) = N_{seed}(t) \tag{25}$$

$$N_{empty}(t) = N_{empty}(t') - \Delta I(t) \times \frac{N_{empty}(t')}{N_{pixel}} \times (1 - e^{-(t-t_i)/\tau_{rec}}). \tag{26}$$

These steps are repeated to calculate $N_{det}(t)$ at each time step. Accordingly, the total numbers of the incident photons and detected photons are:

$$N_{seed} = \sum_{t_i=0}^{t_i=100\ \text{ns}} \Delta I(t_i) \tag{27}$$

$$N_{det} = \sum_{t_i=0}^{t_i=100\ \text{ns}} N_{det}(t_i). \tag{28}$$

Finally, the saturation curve is obtained by scanning the initial number value of the incident photons, $I_0$.

### 3.4. Cross-talk and after-pulse

The effects of correlated noises, such as the so-called cross-talk and after-pulse, are implemented in this model. The cross-talk is a secondary avalanche in a neighboring pixel triggered by infrared photons generated in a primary avalanche. It occurs at the same time as the primary avalanche with a certain probability, $P_{CT}$. Our method counts

[2] We also perform the model calculation with a time step of 0.01 ns and compare the result with that of 0.1 ns. We find that the results are basically the same, yet the calculation for the former (0.01 ns) takes much longer than the latter. Consequently, we conclude that the time step of 0.1 ns is sufficient for our purpose and more suitable in practice.

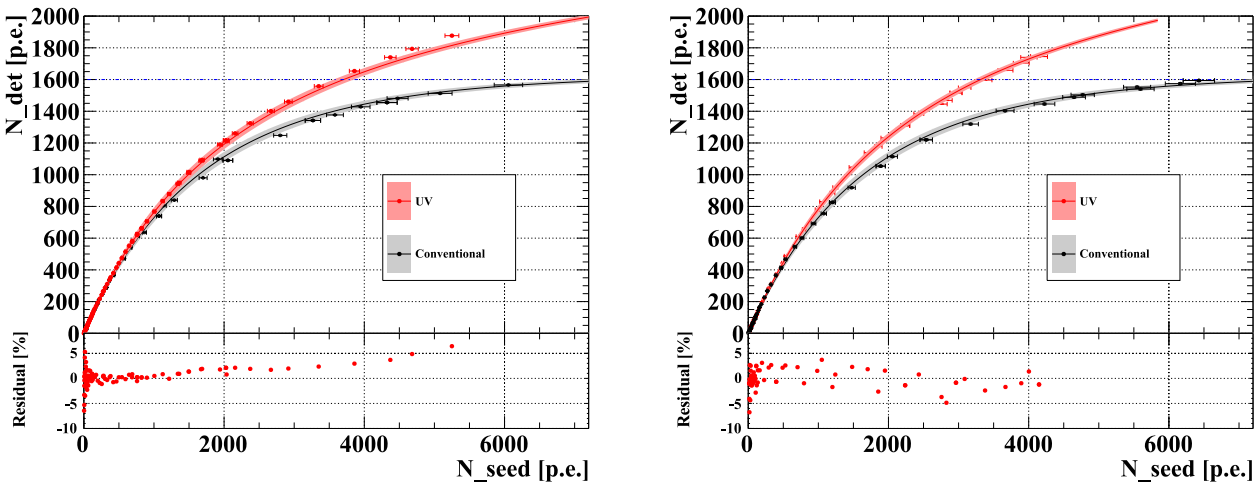


**Fig. 10.** Saturation curves obtained in the measurements and model with (red) UV and (black) visible-light pulse injections. Plots show the measurement results, and the line and belt show the best-fitted model and its error considering the errors of key parameters. $N_{seed}$, $N_{det}$ are the same as in Fig. 9. Bottom figures show the deviation between the measurement and model.

it in the same way as it does the primary signal from scintillation photons and adds it to $N_{det}$.

The after-pulse is a delayed avalanche caused by a trapped carrier in the primary signal and occurs with a certain probability, $P_{AP}$, and a time constant, $\tau_{AP}$.

The complete model, including details of the model implementation of the cross-talk and after-pulse, is described in Appendix.

## 4. Comparison between the measurements and model

This section compares the measurements and the model to verify the new method for SiPM signal saturation measurements. We measure some key parameters with independent measurements using dark noise on the basis of the method suggested by [12]: the recovery time $\tau_{rec} = 6.76 \pm 0.12$ ns, the probability of the cross-talk $P_{CT} = 16.8 \pm 1.1\%$, and that of the after-pulse $P_{AP} = 1.70 \pm 0.02\%$. We choose the values of the time constants of the scintillator such that the saturation curve in the model is best fitted to the measured one. In the fitting, we use the two SiPM measurement data and simultaneously fitted them with $\chi^2/\text{NDF} = 8.27$. The red line in Fig. 10 shows the best-fit saturation model with the optimized time constants of $\tau_{rise} = 0.80 \pm 0.02$ ns and $\tau_{decay} = 2.55 \pm 0.24$ ns, which are nearly consistent with the catalog values of the scintillator in Table 1. In this error estimate, we consider the errors of key parameters of recovery time, cross-talk, and after-pulse probabilities. Consequently, our saturation model is found to well describe the measured saturation curves.

The comparison for the saturation curve measured with visible light is also performed in the same way except that the time distribution of the scintillation light emission is ignored, and the result is overlaid in Fig. 10. The obtained saturation model also matches well with the measured saturation curve.

## 5. Discussion and conclusion

A new method for measuring the saturation of the SiPM in scintillation detectors and its modeling have been developed. The key point of the new method is the use of UV-light pulses as opposed to conventional visible-light pulses to excite scintillation light; with UV-light pulses, the measured saturation curve takes into account the effects of the time constant of the scintillation emission. In the saturation measurement based on the new method, a much larger amount of saturation recovery is observed than that in the conventional method with visible-light pulses. It must be noted that this new method allows in-situ calibration of the SiPM in scintillation detectors. In addition, we have also developed a new saturation model that fully takes into consideration all the major relevant effects for the first time, including the time constant of the scintillation emission, the recovery time of the SiPM pixel, cross-talk, and after-pulse. The measurement and the model are found to be consistent. The new method of saturation measurement and the new saturation model should be useful for accurate saturation correction for various detectors and experiments based on scintillators with SiPM readout.

## CRediT authorship contribution statement

**Naoki Tsuji:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Tatsuki Murata:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Wataru Ootani:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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## Appendix. Model including the cross-talk and after-pulse

### A.1. Cross-talk

For example, at $t = t_0 = 0$ ns,

$$N_{seed}(t_0) = \Delta I(t_0) \tag{A.1}$$

$$\Delta I_{CT}(t_0) = N_{seed}(t_0) \times (P_{CT} + P_{CT}^2 + \cdots + P_{CT}^9) \tag{A.2}$$

$$N_{det}(t_0) = (N_{det} \text{ Calculation using } \Delta I(t_0)) \tag{A.3}$$

$$+ (N_{det} \text{ Calculation using } \Delta I_{CT}(t_0)) \tag{A.4}$$

where $\Delta I_{CT}(t_0)$ is the number of incident photons caused by the cross-talk, and $(P_{CT} + P_{CT}^2 + \cdots)$ considers the second-, third-, $\cdots$, ninth-order effects of the cross-talk.

At $t = t_1 = 0.1$ ns,

$$N_{seed}(t_1) = \Delta I(t_1) \times \frac{N_{empty}(t_0)}{N_{pixel}} \tag{A.5}$$

$$+ \Delta I(t_1) \times \frac{N_{t_0}(t_0)}{N_{pixel}} \times (1 - e^{-(t_1 - t_0)/\tau_{rec}}) \tag{A.6}$$

$$\Delta I_{CT}(t_1) = N_{seed}(t_1) \times (P_{CT} + P_{CT}^2 + \cdots) \tag{A.7}$$

$$N_{det}(t_2) = (N_{det} \text{ Calculation using } \Delta I(t_1)) \tag{A.8}$$

$$+ (N_{det} \text{ Calculation using } \Delta I_{CT}(t_1)) \tag{A.9}$$

The recursive relation with $t$ and $t' = t - 0.1$ is described as:

$$N_{seed}(t) = \Delta I(t) \times \frac{N_{empty}(t')}{N_{pixel}} \tag{A.10}$$

$$+ \sum_{t_i=0}^{t_i=t'} \Delta I(t) \times \frac{N_{t_i}(t')}{N_{pixel}} \times (1 - e^{-(t-t_i)/\tau_{rec}}) \tag{A.11}$$

$$N_{det}(t) = \Delta I(t) \times \frac{N_{empty}(t')}{N_{pixel}} \tag{A.12}$$

$$+ \sum_{t_i=0}^{t_i=t'} \Delta I(t) \times \frac{N_{t_i}(t')}{N_{pixel}} \times (1 - e^{-(t-t_i)/\tau_{rec}})^2 \tag{A.13}$$

$$N_{empty}(t) = N_{empty}(t') \tag{A.14}$$

$$- \Delta I(t) \times \frac{N_{empty}(t')}{N_{pixel}} \times (1 - e^{-(t-t_i)/\tau_{rec}}) \tag{A.15}$$

$$N_{t_i}(t) = N_{t_i}(t') - \Delta I(t) \times \frac{N_{t_i}(t')}{N_{pixel}} \times (1 - e^{-(t-t_i)/\tau_{rec}}) \tag{A.16}$$

$$\Delta I_{CT}(t) = N_{seed}(t) \times (P_{CT} + P_{CT}^2 + \cdots) \tag{A.17}$$

$$N_{seed}(t) = N_{seed}(t) + \Delta I_{CT}(t) \times \frac{N_{empty}(t)}{N_{pixel}} \tag{A.18}$$

$$+ \sum_{t_i=0}^{t_i=t'} \Delta I_{CT}(t) \times \frac{N_{t_i}(t)}{N_{pixel}} \times (1 - e^{-(t-t_i)/\tau_{rec}}) \tag{A.19}$$

$$N_{det}(t) = N_{det}(t) + \Delta I_{CT}(t) \times \frac{N_{empty}(t)}{N_{pixel}} \tag{A.20}$$

$$+ \sum_{t_i=0}^{t_i=t'} \Delta I_{CT}(t) \times \frac{N_{t_i}(t)}{N_{pixel}} \times (1 - e^{-(t-t_i)/\tau_{rec}})^2 \tag{A.21}$$

$$N_{empty}(t) = N_{empty}(t) \tag{A.22}$$

$$- \Delta I_{CT}(t) \times \frac{N_{empty}(t)}{N_{pixel}} \times (1 - e^{-(t-t_i)/\tau_{rec}}) \tag{A.23}$$

$$N_{t_i}(t) = N_{t_i}(t) - \Delta I_{CT}(t) \times \frac{N_{t_i}(t)}{N_{pixel}} \times (1 - e^{-(t-t_i)/\tau_{rec}}) \tag{A.24}$$

$$N_t(t) = N_{seed}(t) \tag{A.25}$$

### A.2. After-pulse

The after-pulse occurs with a certain probability $P_{AP}$ and a time constant $\tau_{AP}$:

$$N_{AP}(t) = P_{AP} \frac{1}{\tau_{AP}} e^{-t/\tau_{AP}} \tag{A.26}$$

where $N_{AP}(t)$ is the number of photons (avalanches) caused by the after-pulse. The number of photons caused by the after-pulse depends on the number of fired pixels and the time difference from the first photon injection.

For example, at $t = t_0 = 0$ ns, there is no after-pulse. At $t = t_1 = 0.1$ ns,

$$N_{AP}(t_1) = N_{t_0} \times P_{AP} \frac{1}{\tau_{AP}} e^{-(t_1 - t_0)/\tau_{AP}} \tag{A.27}$$

$$N_{t_0}(t_1) = N_{t_0}(t_0) - N_{t_0}(t_0) \times P_{AP} \frac{1}{t_{AP}} e^{-(t_1 - t_0)/\tau_{AP}} \tag{A.28}$$

$$N_{det}(t_1) = (N_{det} \text{ Calculation using } \Delta I(t_1) \& N_{t_0}(t_1)) \tag{A.29}$$

$$+ N_{AP}(t_1) \times (1 - e^{-(t_1 - t_0)/\tau_{rec}})^2 \tag{A.30}$$

Since $N_{AP}(t_1)$ is treated as pixels fired at $t_1$, $N_{t_0}$ is updated by subtracting $N_{AP}(t_1)$. The number of detected photons by after-pulse is also affected by the pixel recovery, so the PDE and gain dependence are considered. Then the total number of detected photons at $t_1$ is sum of the $N_{det}$ calculated by the $\Delta I(t_1)$ using updated $N_{t_0}$, and the $N_{det}$ calculated by the $N_{AP}(t_1)$.

The recursive relation with $t$ and $t' = t - 0.1$ is described as:

$$N_{AP}(t) = \sum_{t_i=0}^{t_i=t'} N_{t_i} \times P_{AP} \frac{1}{\tau_{AP}} e^{-(t-t_i)/\tau_{AP}} \tag{A.31}$$

$$N_{empty}(t) = N_{empty}(t') \tag{A.32}$$

$$N_{t_i}(t) = N_{t_i}(t') - N_{t_i}(t') \times P_{AP} \frac{1}{t_{AP}} e^{-(t-t_i)/\tau_{AP}} \tag{A.33}$$

$$N_{seed}(t) = \Delta I(t) \times \frac{N_{empty}(t)}{N_{pixel}} \tag{A.34}$$

$$+ \sum_{t_i=0}^{t_i=t'} \Delta I(t) \times \frac{N_{t_i}(t)}{N_{pixel}} \times (1 - e^{-(t-t_i)/\tau_{rec}}) \tag{A.35}$$

$$+ \sum_{t_i=0}^{t_i=t'} N_{AP}(t_i) \times (1 - e^{-(t-t_i)/\tau_{rec}}) \tag{A.36}$$

$$N_{det}(t) = \Delta I(t) \times \frac{N_{empty}(t)}{N_{pixel}} \tag{A.37}$$

$$+ \sum_{t_i=0}^{t_i=t'} \Delta I(t) \times \frac{N_{t_i}(t)}{N_{pixel}} \times (1 - e^{-(t-t_i)/\tau_{rec}})^2 \tag{A.38}$$

$$+ \sum_{t_i=0}^{t_i=t'} N_{AP}(t_i) \times (1 - e^{-(t-t_i)/\tau_{rec}})^2 \tag{A.39}$$

$$N_{empty}(t) = N_{empty}(t) - \Delta I(t) \times \frac{N_{empty}(t)}{N_{pixel}} \tag{A.40}$$

$$N_{t_i}(t) = N_{t_i}(t) - \Delta I(t) \times \frac{N_{t_i}(t)}{N_{pixel}} \times (1 - e^{-(t-t_i)/\tau_{rec}}) \tag{A.41}$$

$$N_t(t) = N_{seed}(t) \tag{A.42}$$

### A.3. Model

The new saturation model is completed combining all the effects described in the previous sections. The time step for the calculation can be adjusted according to the SiPM types. In the case of the comparison with the measurement using the S12571-015P, it is set as 100 ns divided by 0.1 ns pitch. The recursive relation with $t$ and $t' = t - 0.1$ is

described as:

$$\Delta I(t) = \sum_{t_i=0}^{t_i=t} I_0 e^{-\frac{(t_i-\mu_0)^2}{2\times\sigma_0^2}} \Delta t \tag{A.43}$$

$$\times \int_t^{t+\Delta t} (e^{-(u-t_i)/\tau_{decay}} - e^{-(u-t_i)/\tau_{rise}}) du \tag{A.44}$$

$$N_{AP}(t) = \sum_{t_i=0}^{t_i=t'} N_{t_i} \times P_{AP} \frac{1}{t_{AP}} e^{-(t-t_i)/\tau_{AP}} \tag{A.45}$$

$$N_{empty}(t) = N_{empty}(t') \tag{A.46}$$

$$N_{t_i}(t) = N_{t_i}(t') - N_{t_i}(t') \times P_{AP} \frac{1}{t_{AP}} e^{-(t-t_i)/\tau_{AP}} \tag{A.47}$$

$$N_{seed}(t) = \Delta I(t) \times \frac{N_{empty}(t)}{N_{pixel}} \tag{A.48}$$

$$+ \sum_{t_i=0}^{t_i=t'} \Delta I(t) \times \frac{N_{t_i}(t)}{N_{pixel}} \times (1 - e^{-(t-t_i)/\tau_{rec}}) \tag{A.49}$$

$$+ \sum_{t_i=0}^{t_i=t'} N_{AP}(t_i) \times (1 - e^{-(t-t_i)/\tau_{rec}}) \tag{A.50}$$

$$N_{det}(t) = \Delta I(t) \times \frac{N_{empty}(t)}{N_{pixel}} \tag{A.51}$$

$$+ \sum_{t_i=0}^{t_i=t'} \Delta I(t) \times \frac{N_{t_i}(t)}{N_{pixel}} \times (1 - e^{-(t-t_i)/\tau_{rec}})^2 \tag{A.52}$$

$$+ \sum_{t_i=0}^{t_i=t'} N_{AP}(t_i) \times (1 - e^{-(t-t_i)/\tau_{rec}})^2 \tag{A.53}$$

$$N_{empty}(t) = N_{empty}(t) - \Delta I(t) \times \frac{N_{empty}(t)}{N_{pixel}} \tag{A.54}$$

$$N_{t_i}(t) = N_{t_i}(t) - \Delta I(t) \times \frac{N_{t_i}(t)}{N_{pixel}} \times (1 - e^{-(t-t_i)/\tau_{rec}}) \tag{A.55}$$

$$\Delta I_{CT}(t) = N_{seed}(t) \times (P_{CT} + P_{CT}^2 + \cdots) \tag{A.56}$$

$$N_{seed}(t) = N_{seed}(t) + \Delta I_{CT}(t) \times \frac{N_{empty}(t)}{N_{pixel}} \tag{A.57}$$

$$+ \sum_{t_i=0}^{t_i=t'} \Delta I_{CT}(t) \times \frac{N_{t_i}(t)}{N_{pixel}} \times (1 - e^{-(t-t_i)/\tau_{rec}}) \tag{A.58}$$

$$N_{det}(t) = N_{det}(t) + \Delta I_{CT}(t) \times \frac{N_{empty}(t)}{N_{pixel}} \tag{A.59}$$

$$+ \sum_{t_i=0}^{t_i=t'} \Delta I_{CT}(t) \times \frac{N_{t_i}(t)}{N_{pixel}} \times (1 - e^{-(t-t_i)/\tau_{rec}})^2 \tag{A.60}$$

$$N_{empty}(t) = N_{empty}(t) - \Delta I_{CT}(t) \times \frac{N_{empty}(t)}{N_{pixel}} \tag{A.61}$$

$$N_{t_i}(t) = N_{t_i}(t) - \Delta I_{CT}(t) \times \frac{N_{t_i}(t)}{N_{pixel}} \times (1 - e^{-(t-t_i)/\tau_{rec}}) \tag{A.62}$$

$$N_t(t) = N_{seed}(t) \tag{A.63}$$

Then, the total number of incident photons by the scintillation emission $N_{seed}$ and detected photons by the SiPM $N_{det}$ are described as:

$$N_{seed} = \sum_{t_i=0}^{t_i=100} \Delta I(t) \tag{A.64}$$

$$N_{det} = \sum_{t_i=0}^{t_i=100} N_{det}(t) \tag{A.65}$$

The saturation curve can be obtained by scanning the initial value of the scintillation emission $I_0$. The new saturation model including all the effects of the time constant of the scintillation emission, recovery time, cross-talk, and after-pulse has been developed.

## References


[1] K. Kotera, W. Choi, T. Takeshita, Describing the response of saturated SiPMs, 2015, arXiv:1510.01102.

[2] N. Tsuji, W. Ootani, L. Liu, K. Yoshioka, Y. Morita, M. Gonokami, Study on saturation of SiPM for scintillator calorimeter using UV laser, JINST 15 (2020) C05052.

[3] Hamamatsu Photonics K.K., S12571 series data sheet, 2023, https://www.hamamatsu.com/resources/pdf/ssd/s12571-010_etc_kapd1044e.pdf. (Accessed 25 December 2023).

[4] Hamamatsu Photonics K.K., Technical note MPPC, 2023, https://www.hamamatsu.com/content/dam/hamamatsu-photonics/sites/documents/99_SALES_LIBRARY/ssd/mppc_kapd9008e.pdf. (Accessed 25 December 2023).

[5] PicoQuant Co.,, 2023, https://www.picoquant.com/. (Accessed 25 December 2023).

[6] Advanced laser diode systems - ALS GbbH, 2023, https://www.nktphotonics.com/products/pulsed-diode-lasers/pilas/ (Accessed 25 December 2023).

[7] Eljen Technology Co.,, 2023, https://eljentechnology.com/. (Accessed 25 December 2023).

[8] Hamamatsu Photonics K.K, 2023, https://www.hamamatsu.com/jp/en/index.html. (Accessed 25 December 2023).

[9] Paul Scherrer Institut, 2023, https://www.psi.ch/en/drs (Accessed 25 December 2023).

[10] Tektronix, 2023, https://jp.tek.com/keithley (Accessed 25 December 2023).

[11] D. Jeans, Modeling the response of a recovering SiPM, 2015, arXiv:1511.06528.

[12] S. Gomi, H. Hano, T. Iijima, S. Itoh, K. Kawagoe, S.H. Kim, T. Kubota, T. Maeda, T. Matsumura, Y. Mazuka, K. Miyabayashi, H. Miyata, T. Murakami, T. Nakadaira, T. Nakaya, H. Otono, E. Sano, T. Shinkawa, Y. Sudo, T. Takeshita, M. Taguchi, T. Tsubokawa, S. Uozumi, M. Yamaoka, H. Yamazaki, M. Yokoyama, K. Yoshimura, T. Yoshioka, Development and study of the multi pixel photon counter, Nucl. Instrum. Methods A 581 (2007) 427–432.